

# PHILANI LEONARD MKALIPI

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## PROFESSIONAL SUMMARY

Final-year BSc Computer Science student at the University of the Witwatersrand with hands-on experience building and deploying full-stack web applications. Strong foundation in backend development, databases, REST APIs, and cloud platforms (Azure), with industry exposure through BBD's Software Development Vacation Work Programme. Comfortable working in Agile teams, contributing to production codebases, and learning new technologies quickly.

## EDUCATION

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**University of the Witwatersrand** – Johannesburg, Gauteng

**Bachelor of Science in Computer Science** | Expected December 2025

- Relevant Coursework: Data Structures & Algorithms, Database Systems, Software Design, Computer Networks, Operating Systems, and Information Systems

## TECHNICAL SKILLS

**Languages:** JavaScript (ES6+), TypeScript, SQL, Python, C#, HTML5, CSS3, Material UI, Tailwind CSS.

**Databases:** PostgreSQL, MySQL, Firebase, MongoDB

**Frameworks & Tools:** Node.js, Express, Git/GitHub, React

**Cloud & Platforms:** Azure, Linux (Ubuntu, Kali), Vercel

**Concepts:** REST APIs, Agile Development, System Design

## PROJECTS

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**Vehicle Telematics & Fleet Management System | React, TypeScript, Leaflet.js, PostgreSQL | 2025**

Designed and built a full-stack vehicle tracking dashboard mirroring core industry functionalities.

- Frontend Engineering: Developed a responsive SPA using React and TypeScript, utilizing Material UI for a professional, data-dense dashboard layout.
- Map Integration: Implemented Leaflet.js to visualize vehicle locations, status, and movement on an interactive OpenStreetMap layer.
- Geocoding: Integrated OpenStreetMap Nominatim API to automatically convert user-inputted addresses into precise latitude/longitude coordinates.
- State Management: Managed real-time application state for active vehicle alerts (speeding, offline status) and live statistics.
- Live Demo: <https://vehicle-tracker-mauve.vercel.app/>
- GitHub repository: [https://github.com/Philani101/vehicle\\_tracker](https://github.com/Philani101/vehicle_tracker)

**Community Sports Facility Management Website | University Project | 2025**

- Developed a web application using **vanilla JavaScript**, **PostgreSQL**, and **Azure cloud deployment**.

- Integrated clean CSS styling for responsive UI and implemented Firebase authentication for secure access.
- Link to project website: <https://sportx-dgd3ccama7ahh5aq.canadacentral-01.azurewebsites.net/>
- GitHub repository: [https://github.com/Shar-06/ProjectX\\_Dev\\_Hub](https://github.com/Shar-06/ProjectX_Dev_Hub)

#### Chat Web App | Personal Project | 2025

- Built a real-time chat application with Node.js, Express, and Socket.io that allows users to join chat rooms and communicate instantly with other users.
- Link to project website: <https://word-xchange.onrender.com>
- GitHub repository: <https://github.com/Philani101/Chat-App>

#### SolarWay Suppliers Repairs Management System | Information Systems Project | 2024

*Tools:* C#, Power BI, SQL, Excel, Visual Studio

- Collaborated in a team (HexaTech) to design and develop a **web-based internal repairs management system** for **SolarWay Suppliers**, a Johannesburg-based solar equipment company.
- Conducted **business process analysis**, identifying inefficiencies in manual Excel-based repair tracking and customer communication.
- Designed the system architecture using **MVC pattern** and developed functional modules in **C# (Visual Studio)** for repair tracking, technician assignment, and reporting.
- Implemented a **data analytics and reporting dashboard** in **Power BI and Excel** to provide management with real-time insights into repair trends, technician performance, and turnaround times.
- Delivered comprehensive **system documentation, risk analysis, and feasibility studies** to ensure alignment with business goals and operational needs.
- GitHub repository: <https://github.com/Philani101/Information-Systems-SolarWay>

## EXPERIENCE

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#### Study Trust Mentor | Wits University | Feb 2023 – Oct 2024

- I worked as a Study Trust mentor to 1st Year Undergraduate Wits Students.
- I was responsible for helping students transition from high school to university life.

#### Matric Student Tutor | Phendulani Senior Secondary School | July 2023 (2 week programme)

- Assisted matric students in Mathematics and Science subjects during the winter holidays in preparation for their final exams.

#### Software Development Vacation Work | BBD, Rosebank | 1 Dec 2025 – 12 Dec 2025

- Selected for BBD's in-person vacation work programme, gaining industry exposure to professional software engineering practices.
- Working in a project team on an **Augmented Reality (AR) marketing web application**, contributing to both front-end and back-end components.
- Using **HTML, CSS, TypeScript**, and **Node.js/Express** to build and maintain features for web-based systems.

- Applying secure coding practices (e.g., avoiding unsafe evaluation of user input) and modern development standards.
- Collaborating through Git/GitHub, participating in code reviews, and following Agile rituals including standups, task tracking, and project planning.
- Completing daily timesheets and documenting work progress for university sign-off.
- Working with existing and greenfield codebases, improving maintainability, debugging, and adding new features.